

Assignment – 6

Question – 1

Code:

```
1 // To read a 10-element integer array and print the array.
2
3 #include <stdio.h>
4 int main() {
5     int a[10];
6     printf("Enter 10 integers:\n");
7     for(int i=0; i<10; i++) scanf("%d", &a[i]);
8     printf("\nThe 10 integers are: ");
9     for(int i=0; i<10; i++) printf("%d ", a[i]);
10    printf("\n");
11 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_6$ ./q1
Enter 10 integers:
1 2 3 4 5 7 9 0 12 13
The 10 integers are: 1 2 3 4 5 7 9 0 12 13
```

Question – 2

Code:

```
1 /* Read a 10-element integer array and search for a user given
2 number in the array. The index of the user given number must be
3 printed on the screen as output. */
4
5 #include <stdio.h>
6 int main() {
7     int a[10], num, index = 0;
8     printf("Enter 10 integers: ");
9     for(int i=0; i<10; i++) scanf("%d", &a[i]);
10    printf("The 10 integers are: ");
11    for(int i=0; i<10; i++) printf("%d ", a[i]);
12    printf("\nEnter the number you want to search for: ");
13    scanf("%d", &num);
14    for(int i=0; i<10; i++) {
15        if(num == a[i]) {
16            printf("The index of the element is %d.\n", i+1);
17            index = i+1;
18        }
19    }
20    if(index==0) printf("The number doesn't exist in the array.\n");
21 }
```

Output:

```
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_6$ ./q2
Enter 10 integers: 1 2 3 4 5 6 7 8 9 10
The 10 integers are: 1 2 3 4 5 6 7 8 9 10
Enter the number you want to search for: 6
The index of the element is 6.
students@cseblabadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_6$ ./q2
Enter 10 integers: 1 2 3 4 5 6 7 8 9 10
The 10 integers are: 1 2 3 4 5 6 7 8 9 10
Enter the number you want to search for: 35
The number doesn't exist in the array.
```

Question – 3

Code:

```
1/* To read a 10-element array of real numbers and to find mean,
2variance, and standard deviation of the elements. The mean,
3variance and standard deviation must be printed as output. */
4
5#include <stdio.h>
6#include <math.h>
7
8int main() {
9    int a[10];
10    float mean, var, sum=0, varsum=0;
11    printf("Enter 10 integers:\n");
12    for(int i=0; i<10; i++) scanf("%d", &a[i]);
13    for(int i=0; i<10; i++) sum+=a[i];
14    mean = sum/10;
15    for(int i=0; i<10; i++) varsum += ((a[i]-mean)*(a[i]-mean));
16    var = varsum/9;
17    printf("The mean, variance, and standard deviation of given array is: %.2f, %.3f, %.3f respectively.\n", mean, var, sqrt(var));
18 }
```

Output:

```
students@cselab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_6$ ./q3
Enter 10 integers:
1 2 3 4 5 6 7 8 9 10
The mean, variance, and standard deviation of given array is: 5.50, 9.167, 3.028 respectively.
```

Question – 4

Code:

```
1// To take a 10-element array and find the maximum value stored in the array
2
3#include <stdio.h>
4int main() {
5    int a[10], max;
6    printf("Enter 10 integers:\n");
7    for(int i=0; i<10; i++) scanf("%d", &a[i]);
8    printf("\nThe 10 integers are: ");
9    for(int i=0; i<10; i++) printf("%d ", a[i]);
10    max = a[0];
11    for(int i=1; i<10; i++) {
12        if(max < a[i]) max = a[i];
13    }
14    printf("\nThe maximum value stored in the array is: %d.\n", max);
15 }
```

Output:

```
students@cselab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_6$ ./q4
Enter 10 integers:
1 2 3 5 7 9 10 13
46
79
The 10 integers are: 1 2 3 5 7 9 10 13 46 79
The maximum value stored in the array is: 79.
```

Question – 5

Code:

```
1 /* Take a 2x3 two-dimensional array. Read the elements of this array
2 from user. Create another two-dimensional array to store the
3 transpose of this array. Print both the arrays in matrix form. */
4
5 #include <stdio.h>
6 int main() {
7     int a[2][3], b[3][2];
8     printf("Enter a 2x3 matrix:\n");
9     for(int i=1; i<3; i++) {
10         for(int j=1; j<4; j++) {
11             scanf("%d", &a[i][j]);
12             b[j][i] = a[i][j];
13         }
14     }
15     printf("The given matrix is:\n");
16     for(int i=1; i<3; i++) {
17         for(int j=1; j<4; j++) printf("%d ", a[i][j]);
18         printf("\n");
19     }
20     printf("The transpose matrix is:\n");
21     for(int i=1; i<4; i++) {
22         for(int j=1; j<3; j++) printf("%d ", b[i][j]);
23         printf("\n");
24     }
25 }
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_05$ ./q5
Enter a 2x3 matrix:
1 2 6
9 5 7
The given matrix is:
1 2 6
9 5 7
The transpose matrix is:
1 9
2 5
6 7
```

Question – 6

Code:

```
1 // Take a 3x3 matrix from user. Test whether the matrix is a diagonal matrix.
2
3 #include <stdio.h>
4 int main() {
5     int a[3][3];
6     printf("Enter a 3x3 matrix:\n");
7     for(int i=1; i<4; i++) {
8         for(int j=1; j<4; j++) scanf("%d", &a[i][j]);
9     }
10    printf("The given matrix is:\n");
11    for(int i=1; i<4; i++) {
12        for(int j=1; j<4; j++) printf("%d ", a[i][j]);
13        printf("\n");
14    }
15    if(a[1][2] == 0 && a[1][3] == 0 && a[2][1] == 0 && a[2][3] == 0 && a[3][1] == 0 && a[3][2] == 0 && a[1][1] != 0 && a[2][2] != 0 && a[3][3] != 0) printf("\nThe given matrix is
diagonal.\n");
16    else printf("\nThe given matrix is not diagonal.\n");
17 }
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_05$ ./q6
Enter a 3x3 matrix:
1 0 0
0 5 0
0 0 7
The given matrix is:
1 0 0
0 5 0
0 0 7
The given matrix is diagonal.
```

Question – 7

Code:

```
1 // Take a 3x3 matrix from user. Test whether the matrix is a symmetric matrix.
2
3 #include <stdio.h>
4 int main() {
5     int a[3][3];
6     printf("Enter a 3x3 matrix:\n");
7     for(int i=1; i<4; i++) {
8         for(int j=1; j<4; j++) scanf(" %d", &a[i][j]);
9     }
10    printf("The given matrix is:\n");
11    for(int i=1; i<4; i++) {
12        for(int j=1; j<4; j++) printf("%d ", a[i][j]);
13        printf("\n");
14    }
15    if((a[1][2] == a[2][1]) && (a[1][3] == a[3][1]) && (a[2][3] == a[3][2])) printf("\nThe given matrix is symmetric.\n");
16    else printf("\nThe given matrix is not symmetric.\n");
17 }
```

Output:

```
students@cse1abiadnin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_6$ ./q7
Enter a 3x3 matrix:
1 3 5
5 8 5
9 6 8
The given matrix is:
1 3 5
5 8 5
9 6 8
The given matrix is not symmetric.
```